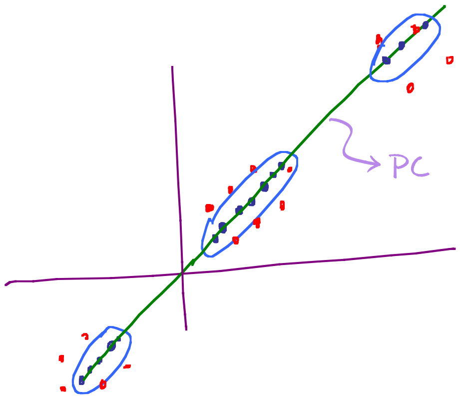
Note Title 8/23/2022

WEEK-III

Unsupervised Learning

- Representation learning
  - PCA
  - kernel PCA
- Clustering (today)



"Clustered together"

Goal: $\{x_1, \dots, x_n\}$ $x_i \in \mathbb{R}^d$

Partition the data into k different clusters

Example: $\{x_1, x_2, x_3, x_4, x_5\}$ $k = 3$

option **1** option **2** option **3**

$\boxed{x_1, x_2, x_5}$ $\boxed{x_3}$ $\boxed{x_4}$

$\boxed{x_1, x_4}$ $\boxed{x_2}$ $\boxed{x_3, x_5}$

Q. How many ways to partition 5 different points into 3 different boxes?

3^5 possibilities

- Number of ways to partition $\mathbf{n}$ different data points into $\mathbf{k}$ different clusters -

k^n

Number of data points

Number of clusters

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$x_1, x_2, \dots, x_n \leftarrow$ DATA POINTS

$z_1, z_2, \dots, z_n \leftarrow$ CLUSTER INDICATORS $z_i \in \{1, \dots, k\}$

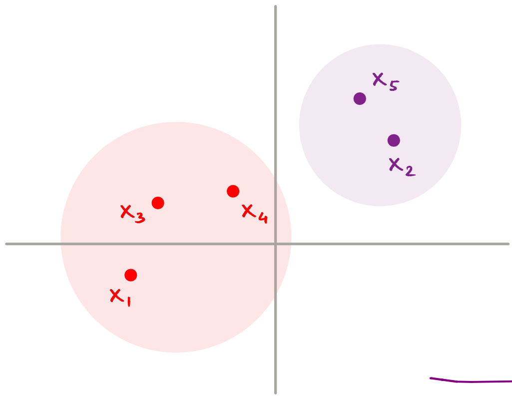
Question: Given a cluster assignment, how good is it?

Objective function: $F(z_1, \dots, z_n) = \sum_{i=1}^{n} \| x_i - \mu_{z_i} \|_2^2$

$\mu_{z_i} \to$ Mean/average of z_i^{th} cluster

$$\mu_k = \frac{\sum_{i=1}^{n} x_i \mathbb{1}(z_i = k)}{\sum_{i=1}^{n} \mathbb{1}(z_i = k)}$$

$$\mathbb{1}(u) = \begin{cases} 1 & \text{if } u \text{ is true} \\ 0 & \text{o/w} \end{cases}$$



Example: x_1 x_2 x_3 x_4 x_5 $k=2$

$z_1=1$ $z_2=2$ $z_3=1$ $z_4=1$ $z_5=2$

$$\mu_1 = \frac{x_1+x_3+x_4}{3} \; ; \; \mu_2 = \frac{x_2+x_5}{2}$$

Goal

$$\min_{\{z_1,\dots,z_n\}} \sum_{i=1}^{n} \|x_i - \mu_{z_i}\|^2$$

Too many possibilities! (k^n)

Upper bound, as it takes empty partitions into account

- Going over all possbile partitions and checking which one gives the least objective value.

$\boxed{\text{NP - HARD}}$

Non-deterministic **P**olynomial Time

- **NP-Hard:** It is not expected that we will get an algorithm to solve this problem whose runtime is polynomial in the parameters of interest, i.e., $\mathbf{n}$ and $\mathbf{k}$.

- Lloyd's algorithm is deterministic. For any fixed initial assignment, it will always yield the same final clusters, provided that ties for equidistant points are resolved consistently.

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LLYOD'S ALGORITHM / K-MEANS ALGORITHM

- Putting **n** different data points into **k** different clusters uniformly at random.

INITIALIZATION

ITERATION

$$z_1^0, z_2^0, \ldots, z_n^0 \in \{1, \ldots, k\}$$

UNTIL CONVERGENCE $\Longrightarrow$ The assignments of data points to clusters no longer change

- COMPUTE MEANS $\quad \forall k \quad \mu_k^t = \dfrac{\sum_{i=1}^{n} x_i \mathbb{1}(z_i^t = k)}{\sum_{i=1}^{n} \mathbb{1}(z_i^t = k)}$

- RE-ASSIGNMENT STEP $\quad \forall i \quad z_i^{t+1} = \arg\min_k \| x_i - \mu_k^t \|_2^2$

[if the mean of current assignment is smallest, then don't reassign]

- No need to reassign a data point if its distance to a new centroid equals the distance to its current centroid.

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FACT: LLYOD'S ALGORITHM CONVERGES. ← Good news.

- Converged solution may not be "optimal"
- But produces "reasonable" clusters in practice.

QUESTIONS

- CONVERGENCE?
- NATURE OF CLUSTERS?
- INITIALIZATION?
- CHOICE OF K?

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LLYOD'S ALGORITHM

QUESTIONS:

1. CONVERGENCE
2. NATURE OF CLUSTERS
3. INITIALIZATION
4. CHOICE OF K.

CONVERGENCE

- Does Llyod's Algorithm Converge? YES

PROOF:

FACT 1: Let $x_1, x_2, \ldots, x_\ell \in \mathbb{R}^d$

$$v^* = \underset{v \in \mathbb{R}^d}{\arg\min} \sum_{i=1}^{\ell} \| x_i - v \|^2$$

[view this objective as a function of v, take derivative, set to 0 and solve]

Answer: $v^* = \frac{1}{\ell} \sum_{i=1}^{n} x_i$

- $\mathbf{v}^*$ is a vector that minimizes the total sum of squared distances to all the given data points.

$\Downarrow$

$\mathbf{v}^*$ is the mean/center of all the given data points.

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- Say we are at iteration t of Llyod's algorithm.

- CURRENT ASSIGNMENT.

Cluster indicator

Iteration number

$$Z_1^t, Z_2^t, \ldots, Z_n^t \in \{1, \ldots, k\}$$

Data point

$\mu_k^t \leftarrow$ Mean of cluster k in iteration t

- Say we update our assignments to

$$Z_1^{t+1}, Z_2^{t+1}, \ldots Z_n^{t+1} \in \{1, \ldots, k\}$$

Not the objective value of the partition that one would get in the $(t+1)^{th}$ round

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As the algorithm has not converged yet,

$$\boxed{\sum_{i=1}^{n} \| x_i - \mu^{t}_{z_i^{t+1}} \|^2} < \boxed{\sum_{i=1}^{n} \| x_i - \mu^{t}_{z_i^{t}} \|^2}$$

[By Algorithmic Choice]

Mean of cluster where x_i wants to go to

$F(z_1^t, \dots, z_n^t)$

[Objective function in the t^{th} iteration]

Mean of current cluster where x_i is assigned to.

$$\sum_{i=1}^{n} \| x_i - \mu^{t+1}_{z_i^{t+1}} \|^2 \leq \underbrace{\sum_{i=1}^{n} \| x_i - \mu^{t}_{z_i^{t+1}} \|^2}_{\text{Intermediate quantity}}$$

$F(z_1^{t+1}, \dots, z_n^{t+1})$

[Objective function in the $(t+1)^{th}$ iteration]

Contribution of each of the data points to the objective function is its distance to the current centroid

Contribution of each of the data points to the objective function was its distance to the previous centroid

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$$\sum_{i=1}^{n} \| x_i - \mu_{z_i^{t+1}}^{t+1} \|^2 = \sum_{i=1}^{n} \underbrace{\sum_{k=1}^{K} \| x_i - \mu_k^{t+1} \|^2 \cdot \mathbb{1}(z_i^{t+1} = k)}$$

For every k

$$\sum_{i \in C_k} \| x_i - \mu_k^{t+1} \|^2 \leq \sum_{i \in C_k} \| x_i - v \|^2$$

any vector

[FACT 1]

$$\leq \sum_{i \in C_k} \| x_i - \boxed{\mu_k^t} \|^2$$

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- The objective value strictly reduces implies the partition cannot be revisited whenever re-assignment happens.

$\Rightarrow$ The objective function strictly reduces after each re-assignment.

$$F(z_1^{t+1}, \dots z_n^{t+1}) < F(z_1^t, \dots, z_n^t)$$

$\Rightarrow$ There are only "FINITE" number of assignments

$\Downarrow$

"FINITE" number of partitions

$\Rightarrow$ Algorithm must converge!

Finiteness does not imply that the algorithm would take $\mathbf{K}^n$ rounds to converge

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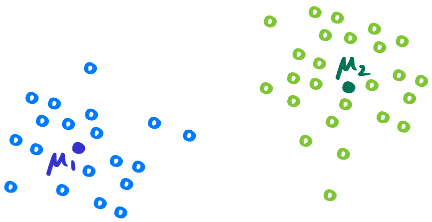
NATURE OF CLUSTERS AFTER CONVERGENCE

<u>K=2</u>



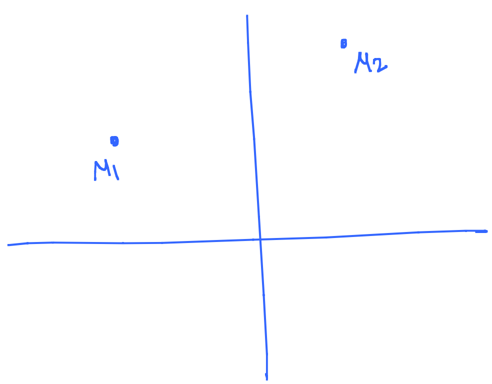
Llyod's algorithm produces 2 clusters with means μ_1 and μ_2

What can we say about points assigned to cluster 1 ?

By Algorithm's construction

For points in Cluster 1,

$$\|x-\mu_1\|^2 \le \|x-\mu_2\|^2$$



$$\|\cancel{x}\|^2 + \|\mu_1\|^2 - 2x^T\mu_1 \le \|\cancel{x}\|^2 + \|\mu_2\|^2 - 2x^T\mu_2$$

$\forall$ x in cluster 1,

$$\boxed{x^T\underbrace{(\mu_2-\mu_1)}_{w} \le \underbrace{\frac{\|\mu_2\|^2 - \|\mu_1\|^2}{2}}_{b}}$$

$$x^T w \le b$$

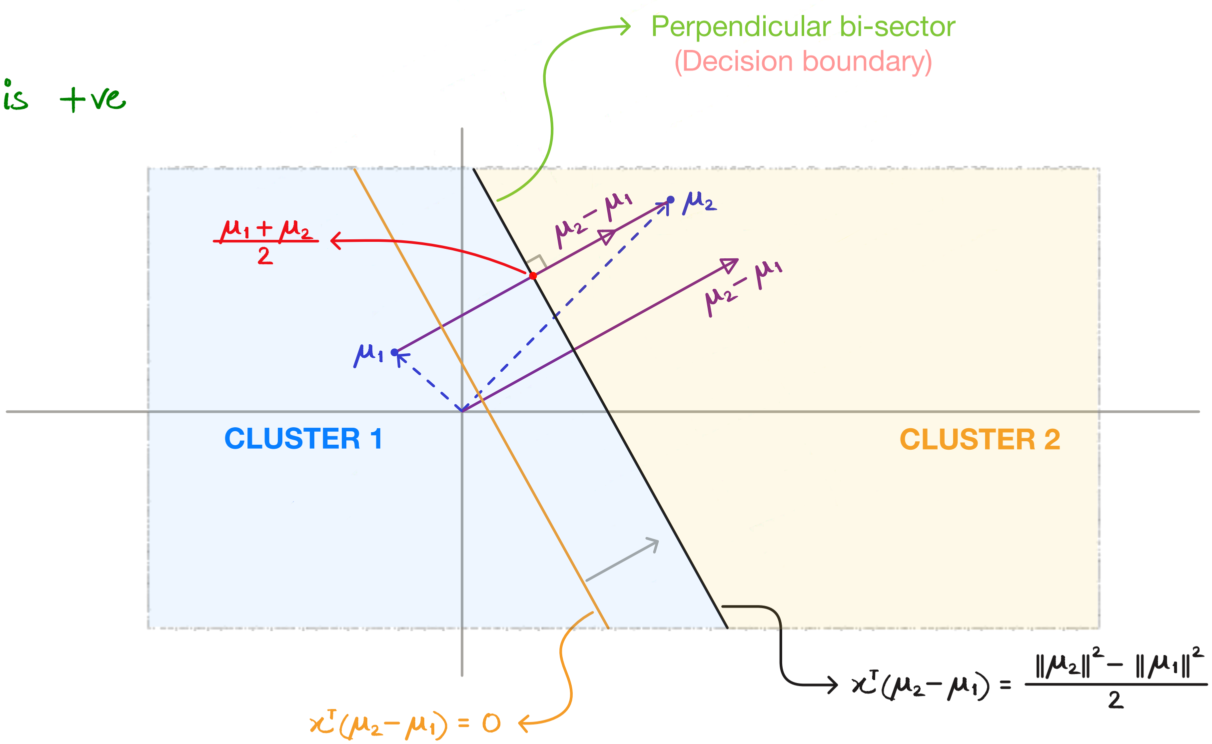
- In this case, $\|\mu_2\|^2 > \|\mu_1\|^2$

$$\Downarrow$$

$\|\mu_2\|^2 - \|\mu_1\|^2$ is $+ve$

$K = 2$

$$x^T(\mu_2-\mu_1) \le \underbrace{\frac{\|\mu_2\|^2 - \|\mu_1\|^2}{2}}_{+ve}$$

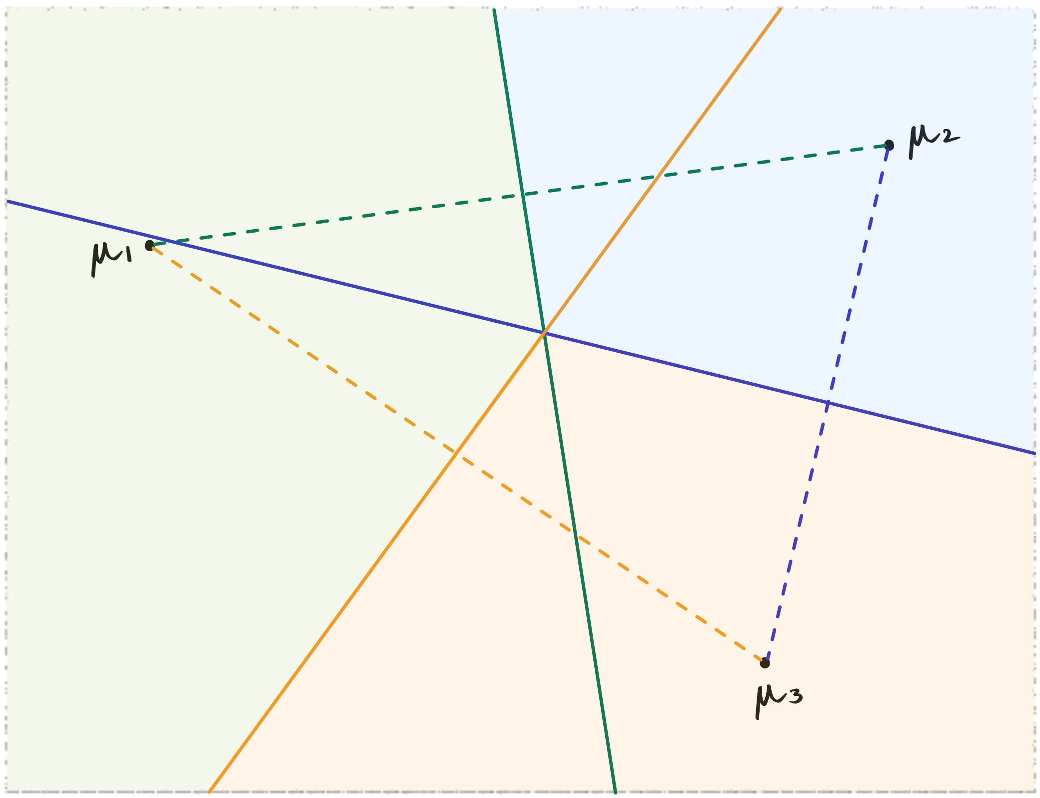


$$x = \frac{\mu_1+\mu_2}{2}$$

$$\left(\frac{\mu_1+\mu_2}{2}\right)^T(\mu_2-\mu_1)$$

$$= \frac{\|\mu_2\|^2 - \|\mu_1\|^2}{2}$$

K = 3



Cluster regions are intersection of half spaces

$\Downarrow$

VORONOI REGIONS

- We are dividing the entire space into **K** different regions using the intersections of half spaces.
- For every cluster, its region would be the intersection of (**K-1**) half spaces.

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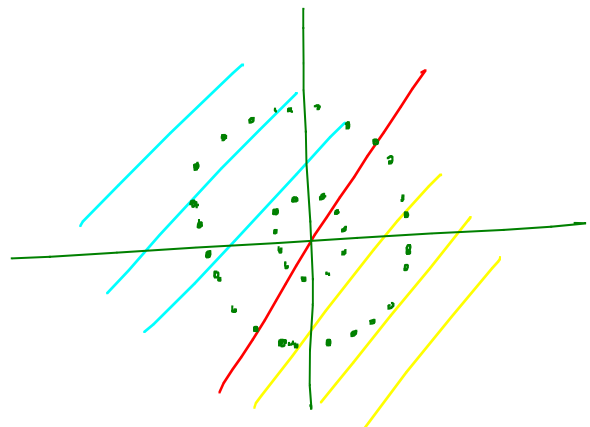

How to fix?

- KERNELIZE K-MEANS
- SPECTRAL CLUSTERING.

- Linear Voronoi regions in the original low-dimensional feature space fail to capture the natural, non-linear clustering of the given dataset. However, if the data is mapped to a high-dimensional transformed feature space, these natural clusters separate out into distinct Voronoi regions, allowing us to successfully run Lloyd's Algorithm in that high-dimensional space.

- The algorithm converges no matter how we initialise it. However, some initialisations might lead us to better clusters. As a result, it is possible to prevent the algorithm from converging at sub-optimal local minima (where the algorithm might get stuck), resulting in higher-quality clusters.

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INITIALIZATION → Instead of initialising the cluster indicators for each data point, we can initialise $\mathbf{k}$ different means

POSSIBILITIES

Random initialisation: Pick K- means uniformly at random from the dataset

- In practice,
  1. Randomly pick K different means/ centroids from the dataset.
  2. Run Lloyd's Algorithm. It'll converge to some partition.
  3. Repeat steps 1 & 2 few more times.
  4. Pick the clustering that gives the least objective value.

K-MEANS ++ → K-Means with a better initialisation

slightly more principled way → Probabilistic initiatisation: Choose first mean μ_1^0 uniformly at random from $\{x_1, \ldots, x_n\}$

Iteration number

- **IDEA:** Centroids should be as far away as possible from each other.

- Initialisation of **K-Means++** is computationally more expensive than **K-Means**.

Note Title 8/29/2022

for $\ell = 2, \dots, k$

Choose μ_ℓ^0 probabilistically proportional to score

$$\boxed{S(x) = \min_{j=1,\dots,\ell-1} \|x - \mu_j^0\|^2} \quad \forall x$$

$x_1, x_2, \boxed{x_3}$ Remaining data points

$10 \quad 20 \quad \boxed{30}$ Scores

$1/6 \quad 1/3 \quad \boxed{1/2}$ Normalising Scores to get a Probability distribution

- The probabilistic aspect of the algorithm provides an expected guarantee of optimal convergence.

GUARANTEE

$$\mathbb{E}\left[\sum_{i=1}^{n} \|x_i - \mu_{z_i}\|^2\right] \leq O(\log k)\left[\min_{z_1,\dots,z_n} \sum_{i=1}^{n} \|x_i - \mu_{z_i}\|^2\right]$$

Big-oh

Over randomness of algorithm

Objective value of the partition Lloyd's algorithm results in. This value is a random quantity as the initialisation was random. The algorithm itself is deterministic.

CHOICE OF K

$\rightarrow$ Hyperparameter

$$\rightarrow \quad F(z_1, \ldots, z_n) = \underbrace{\sum_{i=1}^{n} \|x_i - \mu_{z_i}\|^2}$$

$k = n$

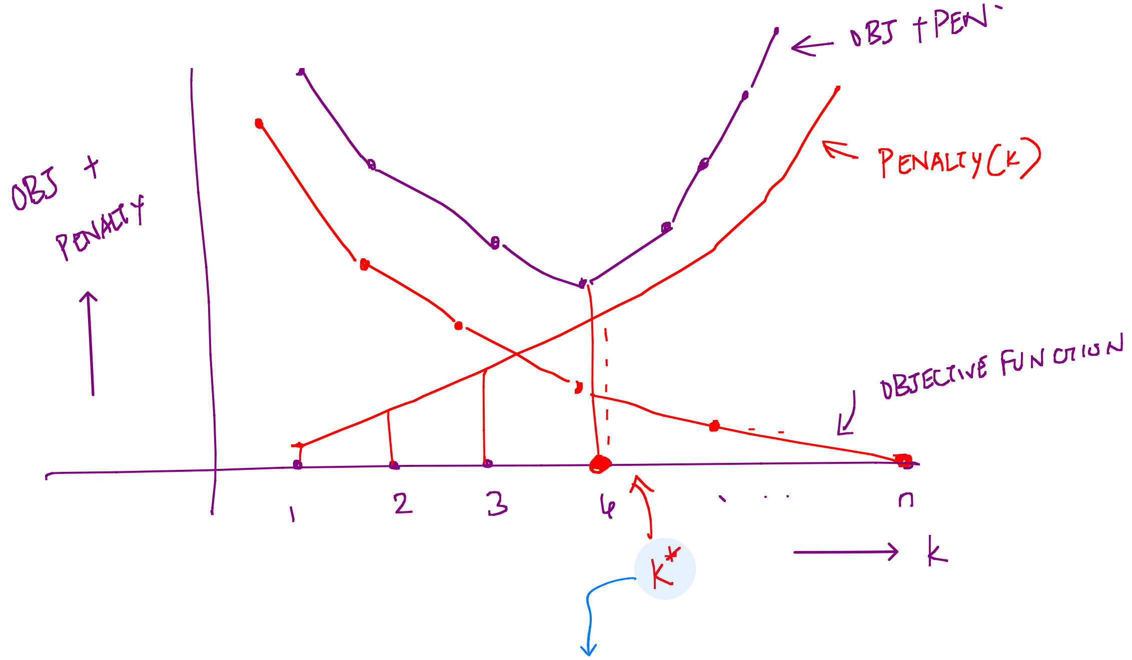
K = **n** gives us the objective value as **0**.

$\rightarrow$ want K to be as small as possible.

$\rightarrow$ Penalize large values of K.

$$\arg\min_k \left(\sum_{i=1}^{n} \|\mathbf{x}_i - \mu_{z_i}\|^2 + \text{Penalty}(k) \right)$$

Find k that has the smallest objective function value + Penalty(k)



Optimum **k** that minimises sum of Objective function and Penalty(**k**)

Some common Criterion for choosing Objective f^n + Penalty

A.I.C - Akaike Information Criterion

$$\left[2K - 2\log\left(L(\theta^*)\right)\right]$$

B.I.C - Bayesian Information Criterion

$$\left[K\log(n) - 2\log\left(L(\theta^*)\right)\right]$$

- CONVERGENCE - YES
- NATURE OF CLUSTERS - VORNOI REGIONS
- INITIALIZATION - K-MEANS++
- CHOICE OF K - OBJ + PENALTY(K).